

Shipping Route Efficiency - Nassau Candy Distributor

Data Analyst Intern · 02 October 2026 · 10,194 shipments, Jan 2024 – Dec 2025

4.29 d avg processing lead time	14.0% delay frequency	7.4% → 18.7% breach rate 2024 → 2025	–49% chocolate miles via nearest plant
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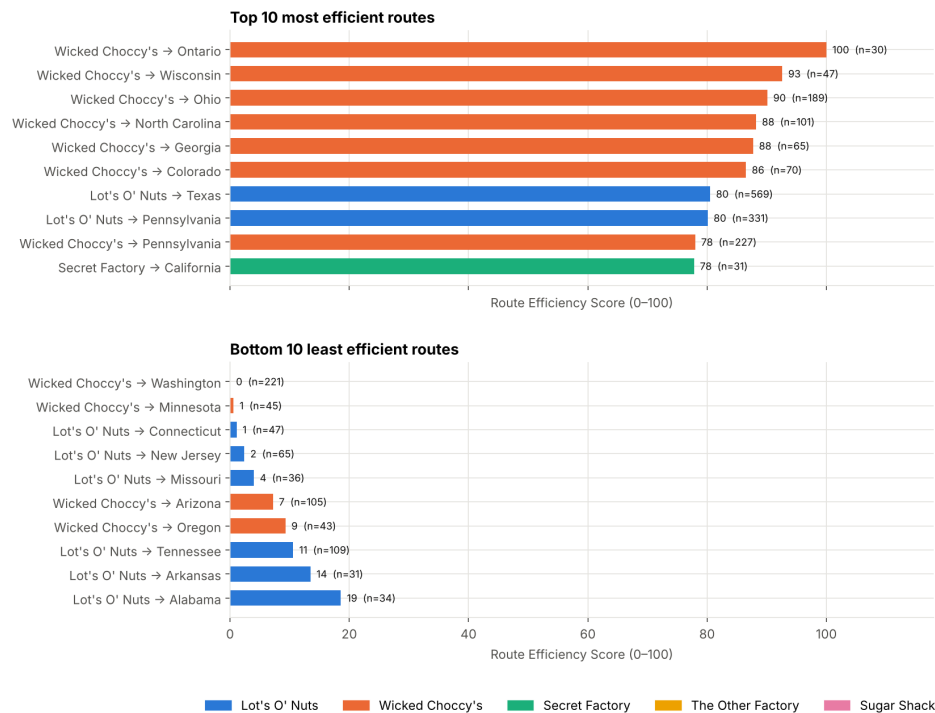
The question. Which factory → customer routes are efficient, where are the bottlenecks, how do ship modes compare, and can the factory network be better aligned with demand?

What we found

- **Delays are an operations problem, not a distance problem.** Processing lead time is set by ship mode and shows no relationship with route distance ($p = 0.01$).
- **Service deteriorated in 2025.** SLA breaches rose from 7.4% to 18.7% – in every ship mode, with a stable mode mix – pointing to a fulfilment-capacity or process change.
- **Premium service is unreliable.** 14.6% of First Class shipments miss their 3-day SLA.
- **Bottleneck states:** Minnesota, Missouri, Utah, Oklahoma, Indiana, Washington are significantly slower than average.
- **Worst routes:** Wicked Choccy's → Washington, Wicked Choccy's → Minnesota, Lot's O' Nuts → Connecticut – long lanes with above-average delays. **Best:** short regional lanes such as Wicked Choccy's → Wisconsin and Wicked Choccy's → Ohio.
- **The network is crossed.** The Arizona plant ships heavily to the East Coast and the Georgia plant to the West Coast.
- **Data issue fixed:** raw ship dates were offset by whole years (904–1,642-day gaps); corrected before analysis.

What we recommend

#	Recommendation	Evidence	Expected impact
1	Root-cause the 2025 dispatch slowdown (staffing, carrier pick-up windows, order cut-off times) and add fulfilment capacity.	SLA breach rose from 7.4% (2024) to 18.7% (2025), inside every ship mode, with a stable mode mix.	Bring breach rate back toward the 2024 level.
2	Dual-source Wonka Bars from the Arizona and Georgia plants, routing each order to the nearer plant.	Cuts chocolate unit-miles by 49% (56% if Secret Factory is added as a third source).	Lower freight cost and real transit time; removes most cross-country lanes.
3	Protect First Class with a dedicated pick lane or earlier cut-off.	14.6% of First Class shipments miss the 3-day SLA – premium customers are let down roughly one time in seven.	Higher premium-service reliability and retention.
4	Targeted review of bottleneck states (Minnesota, Missouri, Utah, Oklahoma, Indiana, Washington, Illinois).	Significantly slower than average after controlling for ship mode (95% CI above zero).	Remove local carrier / hub issues.
5	Fix the date-shift defect in the order/ship-date export and add freight cost and delivery date to the data.	Raw Ship – Order gaps of 904–1,642 days; margins look identical across ship modes.	Trustworthy KPIs and true cost-to-serve monitoring.



Route efficiency leaderboard – top and bottom 10 factory → state routes.